GAZEMM: ROBUST OFF-ANGLE IRIS RECOGNITION VIA GAZE MANIFOLD MATCHING — SUPPLEMENTARY MATERIAL —

Yewon Min[⋆] *Youngwook Kim*[⋆] *Jiyoung Park* *Moonwook Ryu* *Jun Seong Lee*[†]

Electronics and Telecommunications Research Institute
Daejeon, Republic of Korea

This supplementary material provides the technical and experimental details that were streamlined in the main paper due to the page limit. It mainly includes:

(A) Dataset Details

(B) Implementation Details

(C) Additional Experimental Results

Tables 1–4, Figs. 1–2, and Eqs. (1)–(6) refer to items in the main paper. Items of this document carry the prefix "S".

A. DATASET DETAILS

Sec. 3.1 of the main paper lists the twelve datasets. Here we provide an extended description.

- Fig. S1 shows a representative image of each dataset. The eight frontal datasets (CASIA v1.0, CASIA-Iris-Interval, CASIA-Iris-Thousand, CASIA-Iris-Lamp [1], PolyU Iris DB [2], MMU-v1 [3], and the Cogent and Vista subsets of the IIITD Contact Lens Iris DB [4]) are near-infrared benchmarks acquired under cooperative, frontal-gaze conditions. They differ in sensor, resolution, illumination, and pupil dilation, and therefore provide diverse quality distributions.
- The four off-angle datasets are complementary in acquisition conditions. EyeMovement [5] consists of eye-movement videos recorded while users continuously followed gaze-guiding targets on a screen. OpenEDS [6] was captured at close range and off-axis by cameras embedded in a VR headset. CASIA-Iris-Degradation [1] contains controlled subsets for gaze (four directions), illumination, occlusion, and eye state. CASIA-Iris-Africa [7] exhibits diverse gaze, illumination, and occlusion variations.
- Table S1 lists the number of images and eye-level identities used in the experiments and the fraction of images removed by the quality gate of Sec. 2.2.1 in the main paper. Identities are defined at the eye level, i.e., the left and right eyes of the same person are treated as different classes, and only comparisons between the same eye are regarded as genuine.

[⋆]These authors contributed equally to this work. [†]Corresponding author.

Table S1. Datasets used in the experiments. #IDs counts eye-level identities. "Test/fold" is the number of test images in the first fold of the 3×2 cross-validation protocol (Sec. 3.1 of the main paper). "Removed" is the fraction of images removed by the quality gate with $c{=}2$, which is reported as the FTE of the proposed method in Tables 1 and 2 of the main paper.

Dataset	#Images	#IDs	Test/fold	Removed (%)
Frontal				
CASIA v1.0 (V1)	756	108	378	2.91
CASIA-Iris-Interval (Int)	2,639	393	1,382	3.49
CASIA-Iris-Thousand (Tho)	20,000	2,000	10,000	0.10
CASIA-Iris-Lamp (Lam)	16,212	819	8,121	3.97
PolyU Iris DB (Pol)	6,270	417	3,135	3.48
MMU-v1 (MMU)	450	88	225	4.00
IIITD-CLI Cogent (Cog)	1,163	202	579	0.00
IIITD-CLI Vista (Vis)	1,000	198	500	3.30
Off-angle				
EyeMovement (EM)	9,120	76	4,560	1.56
OpenEDS (OED)	27,431	186	13,715	0.36
CASIA-Iris-Degradation (Deg)	36,539	510	18,400	3.08
CASIA-Iris-Africa (Afr)	28,717	2,044	14,217	6.35

Images removed by the quality gate. Fig. S2 shows examples of images removed by the quality gate of Sec. 2.2.1 in the main paper, namely one representative removed image for each of the four off-angle datasets. The removed images are either blurred, so that the iris texture is not resolvable, or captured with the eye closed. Because the gate is a per-dataset statistic, the same rule removes 0.00–6.35% of the images depending on the dataset (Table S1).

B. IMPLEMENTATION DETAILS

Proposed method. The backbone is an ImageNet-pretrained ResNet34 whose classifier is replaced by a 512-dimensional embedding layer. The input is the grayscale periocular image, padded to a square with gray (127) without segmentation, resized to 224×224, replicated to three channels, and normalized to $[-1, 1]$. Each training batch contains $P{=}16$ identities $\times$ $K{=}4$ images (batch size 64), and the network is trained with the soft-margin (softplus) batch-hard triplet loss of Eq. (3) [8] using Adagrad (learning rate 0.075, weight

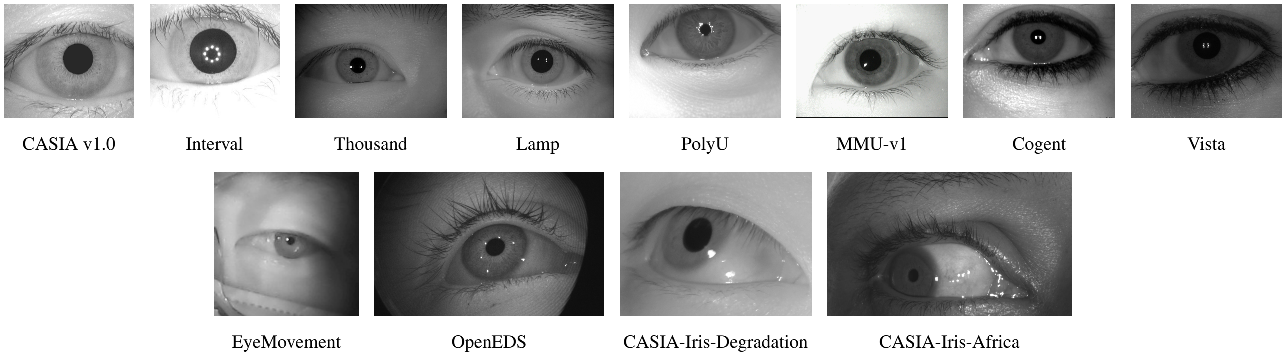

Fig. S1. Representative images of each dataset. Top: the eight frontal datasets. Bottom: the four off-angle datasets, which exhibit perspective distortion due to gaze deviation, loss of iris boundaries, and occlusion and illumination variations.

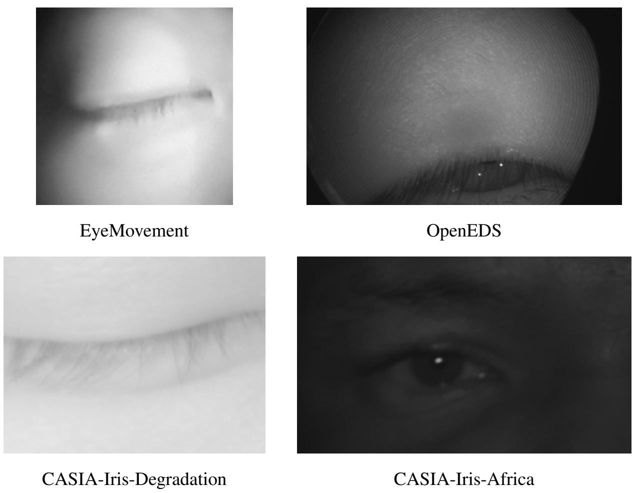

Fig. S2. Representative images removed by the quality gate (c=2), one per off-angle dataset. The removed images are blurred or captured with the eye closed, so that the iris texture is not resolvable.

decay 10^{-5}) for at most 150 epochs. Early stopping on the validation EER (patience 10) selects the checkpoint. The gaze manifold of an identity is the affine subspace spanned by the top k=3 right singular vectors of its mean-removed embeddings (equivalently, the top eigenvectors of $A_i A_i^\top$ in Eq. (4)). The same k, c, and normalization are used for all twelve datasets and for all feature extractors in Table 3 of the main paper. No per-dataset tuning is performed.

Comparison methods. All comparison methods run the authors' released implementations with their default or recommended settings. The individual methods are as follows.

- **WAHET** [9]: a real-time segmentation method that estimates the pupil and iris boundaries with a weighted adaptive Hough transform and an ellipsopolar transform. We run the official USIT v3.0.0 chain `wahet` (segmentation and rubber-sheet normalization to 512×64), `lg` (1-D log-Gabor code), and `hd` (minimum Hamming distance over ±16 bit shifts).
- **USITv3** [10]: the open iris recognition toolkit of the University of Salzburg. We follow the combination recommended by the toolkit authors and used in [11]: `caht` (contrast-adjusted Hough transform), `qsw` (quadratic spline wavelet code [12]), and `hd` with the `-sf` option, which excludes pairs whose masks do not overlap at any shift instead of assigning them a distance of zero.
- **OSIRIS v4.1** [13]: the BioSecure reference implementation with Viterbi-based boundary estimation, Gabor phase encoding, and fractional Hamming distance. It uses the default `process.ini` of the official distribution (pupil diameter 50–160, iris diameter 160–280, 512×64 normalization, official filter bank). The all-pairs matching was vectorized and verified to reproduce the official scores exactly.
- **WCI** [14]: the open pipeline (open-iris) of Worldcoin, consisting of CNN segmentation, classical iris-code encoding, and Hamming-distance matching. We run the complete pipeline including its built-in validators (off-gaze, occlusion, pupil-to-iris ratio, and sharpness thresholds), with the released segmentation weights as-is, and match iris codes with the official `HammingDistanceMatcher` (rotation shift 15, separate half matching, normalized distance).
- **HDBIF** [15]: binarized statistical image features with BSIF filters retrained to be iris-domain-specific, on the CVRL segmentation. The released implementation and filter weights are used with masked fractional Hamming distance. The official code contains no quality check, so none is applied.
- **ArcIris** and **TripletIris** [11]: CNN embeddings trained with the ArcFace loss (ResNet-100) and with a triplet loss (ConvNeXt-tiny), respectively, on normalized iris images. Both are evaluated zero-shot with the released weights. ArcIris calls the official `checkQuality` (minimum pupil and iris radii of 12 and 16 pixels), which is the

source of its FTE. TripletIris has no quality check in the official code.

- **DGR** [16]: represents and matches features as dynamic graphs for robustness to occlusion. It is evaluated zero-shot with the released single-scale weights and the official graph matching. Because its score is a graph similarity rather than a distance between embedding vectors, it cannot be combined with the proposed matching and is therefore absent from Table 3 of the main paper.
- **UE-UGCL** and **Maxout-BA** [17]: an embedding method that jointly learns uncertainty factors, and the baseline maxout CNN of the same repository. No trained weights are released, so both are retrained on the training folds of the same splits as the proposed method with the official settings: inputs normalized by open-iris and resized to 128×128 (UE-UGCL additionally applies the 3×3 median filter of its enhancement module), ArcFace margin softmax (s=64, m=0.5) with the KL term (λ=0.01) and the UGCL curriculum for UE-UGCL, SGD (initial learning rate 0.001, momentum 0.9, weight decay 0.001) with $\times0.1$ decay at epochs 30/50/70, 100 epochs, and batch size 300. Matching uses the cosine distance of L_2-normalized embeddings. Two deviations from the original papers are made for a fair protocol: no external-data pretraining is used, and the checkpoint is selected by the validation EER (as for the proposed method) instead of reporting the last epoch. Their published numbers are therefore not comparable with Tables 1 and 2 of the main paper. DGR, UE-UGCL, and Maxout-BA share the same open-iris normalization stage (without the validators), which is why their FTE values coincide.

Table S2. Per-dataset EER (%) on the eight frontal datasets (abbreviations as in Table S1). **Bold** and underlined: best and second best per column.

Method	V1	Int	Tho	Lam	Pol	MMU	Cog	Vis
WAHET	1.56	1.87	8.98	15.62	24.86	4.40	10.61	6.39
USITv3	0.26	0.67	22.47	18.43	25.00	3.23	13.36	5.60
OSIRIS	0.60	0.89	6.60	4.66	14.87	6.12	1.62	1.63
HDBIF	0.67	0.89	4.13	2.10	1.62	1.34	0.19	0.34
WCI	**0.00**	**0.00**	0.49	0.12	0.79	**0.02**	**0.00**	**0.00**
DGR	2.61	1.65	3.18	4.93	5.02	1.55	2.51	4.05
UE-UGCL	15.93	11.13	7.69	6.06	5.67	12.72	12.69	12.02
Maxout-BA	22.60	17.54	10.59	10.11	9.05	16.49	20.88	19.96
ArcIris	0.29	0.32	1.32	0.84	0.99	1.08	**0.00**	0.35
TripletIris	0.33	0.46	2.43	0.68	1.07	0.58	0.89	0.20
GazeMM (Ours)	0.38	0.22	**0.22**	**0.00**	**0.04**	1.04	0.60	0.28

Table S3. Per-dataset FRR@FAR= 10^{-3} (%) on the eight frontal datasets.

Method	V1	Int	Tho	Lam	Pol	MMU	Cog	Vis
WAHET	8.61	4.73	52.50	86.85	45.10	23.93	20.01	22.43
USITv3	0.26	0.77	46.27	38.38	43.05	5.30	20.57	7.19
OSIRIS	0.97	1.16	21.35	13.61	36.64	15.11	2.48	2.05
HDBIF	1.21	1.41	10.56	4.07	3.22	1.89	0.33	0.42
WCI	**0.00**	**0.00**	0.68	0.14	0.97	**0.00**	**0.00**	**0.00**
DGR	7.34	5.73	14.51	27.59	20.08	5.30	8.43	9.10
UE-UGCL	68.40	56.41	40.95	33.75	26.51	68.56	58.03	55.35
Maxout-BA	81.11	74.49	54.72	49.77	41.45	76.15	75.64	75.13
ArcIris	0.38	0.43	3.25	1.40	1.67	1.74	**0.00**	0.45
TripletIris	1.23	1.61	12.17	2.03	3.02	1.48	3.26	0.37
GazeMM (Ours)	2.14	0.76	**0.54**	**0.00**	**0.02**	6.47	4.21	0.83

C. ADDITIONAL EXPERIMENTAL RESULTS

Per-dataset results on the frontal datasets. Table 1 of the main paper reports only averages over the eight frontal datasets. Tables S2–S4 give the per-dataset EER, FRR@FAR= 10^{-3}, and FTE. The trends match the main paper: on frontal data the strongest methods (WCI, ArcIris, TripletIris, and the proposed method) are within about 1% EER of each other on most datasets, and the proposed method is best or second best on four of the eight datasets. Its largest frontal error occurs on MMU-v1 (EER 1.04%), the smallest dataset (450 images, about five images per identity), where the leave-one-out manifold is estimated from only four images. This is consistent with the analysis of k below. WCI reaches the lowest frontal errors but at the cost of rejecting 33.9% and 84.1% of the PolyU and Vista images by its validators (Table S4), so its scores are computed on a much easier subset.

Per-dataset results on the off-angle datasets. Table 2 of the main paper gives the EER and FTE on the four off-angle datasets. Tables S5 and S6 add the FRR@FAR= 10^{-3} and Rank-1 accuracy. The proposed method has the lowest FRR and the highest Rank-1 on every dataset. On CASIA-Iris-Africa the Rank-1 accuracy of all methods is low (below 80%): the dataset has only about 14 images per eye, acquired under unconstrained conditions with heavy occlusion, so many probes have few usable same-identity counterparts. The verification metrics on this dataset are nevertheless consistent with those of the other three datasets.

Per-dataset generalization across feature extractors. Table 3 of the main paper averages the generalization experiment over the four off-angle datasets. Table S7 gives the per-dataset values. In this experiment, "point-to-point" is each extractor's original pairwise distance (cosine, angular, or Euclidean) without score normalization, whereas the proposed matching comprises both the manifold residual distance and the local-scaling normalization. The point-to-point row of Table 4 in the main paper, in contrast, keeps the normalization. Replacing point-to-point matching by the proposed matching reduces the EER on every dataset by roughly an order of magnitude or more for the four external extractors, and by a factor of 3 to 24 for the proposed embedding, and it lowers the FRR@FAR= 10^{-3} in every case. The gain is

Table S4. Per-dataset failure-to-enroll rate (FTE, %) on the eight frontal datasets.

Method	V1	Int	Tho	Lam	Pol	MMU	Cog	Vis
WAHET	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
USITv3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
OSIRIS	0.0	0.0	0.2	0.1	10.1	0.9	0.3	0.0
HDBIF	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
WCI	1.7	0.9	7.7	2.2	33.9	0.0	2.5	84.1
DGR	0.9	0.8	7.6	1.8	0.7	0.0	0.0	0.0
UE-UGCL	0.9	0.8	7.6	1.8	0.7	0.0	0.0	0.0
Maxout-BA	0.9	0.8	7.6	1.8	0.7	0.0	0.0	0.0
ArcIris	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
TripletIris	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
GazeMM (Ours)	2.9	3.5	0.1	4.0	3.5	4.0	0.0	3.3

Table S5. Per-dataset FRR@FAR= 10^{-3} (%) on the four off-angle datasets (EER and FTE are in Table 2 of the main paper).

Method	EM	OED	Deg	Afr
WAHET	96.36	99.68	99.62	85.46
USITv3	95.44	93.86	59.46	78.08
OSIRIS	95.13	90.85	83.44	62.28
HDBIF	71.75	89.37	54.00	57.97
WCI	63.95	88.38	11.29	19.21
DGR	62.08	88.90	75.64	63.76
UE-UGCL	57.26	87.44	63.92	56.57
Maxout-BA	69.42	90.87	77.78	86.49
ArcIris	67.13	88.21	37.91	44.46
TripletIris	78.42	87.55	90.12	72.45
GazeMM (Ours)	**0.00**	**31.32**	**1.23**	**0.20**

smallest on OpenEDS, where the FRR of the four external extractors remains high (48–61%) and that of the proposed embedding is 31%. OpenEDS is an eye-tracking dataset whose frames include partially closed eyes and strong eyelid occlusion, for which little iris texture is visible in any representation. The absolute errors after manifold matching follow the quality of the underlying embeddings (ArcIris and the proposed ResNet34 embedding are the strongest), which confirms that the gain comes from the matching stage and is independent of the extractor.

Component ablation on all off-angle datasets. Table 4 of the main paper reports the ablation on CASIA-Iris-Degradation. Table S8 repeats it on all four off-angle datasets and adds the fold-to-fold standard deviation. Replacing manifold matching by point-to-point matching is by far the most harmful change on every dataset. Score normalization helps clearly on CASIA-Iris-Degradation, whereas on the other three datasets the remaining differences between the variants are within one standard deviation. The quality gate changes the mean errors only marginally, but it reduces the variance of the strict operating point on CASIA-Iris-Degradation (FRR $2.00 \pm 2.15\% \rightarrow 1.23 \pm 0.53\%$).

Table S6. Per-dataset Rank-1 accuracy (%) on the four off-angle datasets.

Method	EM	OED	Deg	Afr
WAHET	18.26	1.18	0.30	22.66
USITv3	46.77	66.61	58.89	28.18
OSIRIS	43.79	88.36	76.29	39.82
HDBIF	84.52	87.96	87.32	48.42
WCI	99.15	92.81	98.04	58.31
DGR	99.08	93.62	96.34	46.81
UE-UGCL	98.72	92.94	95.84	48.91
Maxout-BA	97.79	90.96	93.76	34.65
ArcIris	98.22	92.57	92.75	52.37
TripletIris	97.45	92.36	91.36	46.68
GazeMM (Ours)	**100.00**	**99.65**	**99.74**	**77.11**

Effect of the subspace dimension k. Table S9 varies k from 1 to 20 on the four off-angle datasets with everything else fixed. Even k=1, i.e., a single dominant gaze direction per identity, already removes most of the point-to-point error (compare with the P2P rows of Table S7), which supports the low-dimensional hypothesis of Sec. 1 in the main paper. EyeMovement is saturated for every k. OpenEDS and CASIA-Iris-Degradation keep improving up to k=20, because these datasets provide many images per identity (about 150 and 70, respectively), so a higher-dimensional subspace can be estimated reliably. CASIA-Iris-Africa, with about 14 images per identity, is best at k=5 and degrades beyond it: as k approaches the number of enrollment images, the subspace absorbs impostor directions as well and the residual loses discriminability. The value k=3 used in the main paper was fixed a priori for all datasets and extractors. It is therefore a conservative choice, and the main-paper results on OpenEDS and CASIA-Iris-Degradation would improve further with a larger k. A data-dependent choice, e.g., k proportional to the number of enrollment images, is left for future work.

Sensitivity to the quality-gate strictness c. Table S10 repeats the full experiment (training included) with $c \in \{2, 3, 4\}$ on all twelve datasets. The gate removes 0–6.4% of the images at c=2, 0–3.1% at c=3, and almost nothing at c=4, yet the EER changes by at most about 0.2 points on every dataset, and not in a consistent direction. The main effect of the gate is on the stability of the strict operating point: on CASIA-Iris-Degradation the fold-to-fold standard deviation of FRR@FAR= 10^{-3} grows from $\pm$0.5 at c=2 to $\pm$2.2 at c=4 (Table S8), because a few severely blurred enrollment frames distort the manifold of their identity. The gate is thus a safeguard rather than a tuned component, and the same c=2 is used everywhere.

Table S7. Per-dataset generalization of the proposed matching across feature extractors on the four off-angle datasets (EER / FRR@FAR= 10^{-3}, %). P2P is the extractor's original point-to-point matching. +GazeMM is the same embeddings matched with the proposed manifold matching and score normalization. Averages over the four datasets appear in Table 3 of the main paper.

Extractor	Matching	EM	OED	Deg	Afr
UE-UGCL	P2P	10.70 / 57.26	11.61 / 87.44	13.16 / 63.92	12.45 / 56.57
	+GazeMM	**0.56 / 1.98**	**1.19 / 51.84**	**0.34 / 0.97**	**0.62 / 1.88**
Maxout-BA	P2P	15.10 / 69.42	27.62 / 90.87	18.64 / 77.78	27.16 / 86.49
	+GazeMM	**1.04 / 5.98**	**1.95 / 61.46**	**1.04 / 6.69**	**1.98 / 11.43**
ArcIris	P2P	15.64 / 67.13	16.20 / 88.21	16.54 / 37.91	17.69 / 44.46
	+GazeMM	**0.32 / 4.77**	**1.07 / 47.78**	**0.10 / 0.10**	**0.50 / 0.78**
TripletIris	P2P	14.94 / 78.42	11.98 / 87.55	17.98 / 90.12	17.60 / 72.45
	+GazeMM	**0.64 / 2.64**	**1.13 / 53.33**	**0.69 / 2.17**	**0.68 / 2.24**
ResNet34 (Ours)	P2P	0.98 / 5.64	1.58 / 61.52	5.11 / 46.26	2.36 / 17.44
	+GazeMM	**0.00 / 0.00**	**0.57 / 31.32**	**0.25 / 1.23**	**0.10 / 0.20**

Table S8. Component-wise ablation on the four off-angle datasets (EER / FRR@FAR= 10^{-3}, %, mean±std over 3×2 folds). The point-to-point variant keeps the score normalization and the quality gate. "w/o quality gate" trains and evaluates with the gate disabled (FTE 0%).

Variant	EM	OED	Deg	Afr
Point-to-point	0.93±0.60 / 5.06±2.17	1.57±0.26 / 61.10±3.91	4.98±0.38 / 45.41±3.06	2.33±0.22 / 17.30±2.71
w/o score norm.	0.03±0.04 / 0.04±0.05	0.63±0.12 / 31.05±7.09	0.60±0.06 / 4.49±1.39	0.12±0.02 / 0.25±0.15
w/o quality gate	0.00±0.00 / 0.00±0.00	0.58±0.09 / 30.27±6.71	0.29±0.09 / 2.00±2.15	0.14±0.03 / 0.70±0.54
GazeMM (full)	0.00±0.00 / 0.00±0.00	0.57±0.07 / 31.32±8.39	0.25±0.02 / 1.23±0.53	0.10±0.02 / 0.20±0.17

Table S9. Effect of the subspace dimension k on the four off-angle datasets (EER / FRR@FAR= 10^{-3}, %, mean over 3×2 folds). The main paper uses k=3 for all datasets and extractors.

k	EM	OED	Deg	Afr
1	0.04 / 0.03	0.73 / 41.84	0.55 / 5.72	0.17 / 1.16
2	0.01 / 0.00	0.63 / 35.51	0.34 / 2.54	0.12 / 0.41
3	0.00 / 0.00	0.57 / 31.32	0.25 / 1.23	0.10 / 0.20
5	0.00 / 0.00	0.49 / 25.87	0.14 / 0.29	0.09 / 0.08
10	0.00 / 0.00	0.39 / 19.36	0.05 / 0.01	0.10 / 0.12
15	0.00 / 0.00	0.35 / 16.55	0.02 / 0.00	0.11 / 0.17
20	0.00 / 0.00	0.31 / 13.77	0.01 / 0.00	0.12 / 0.20

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Table S10. Sensitivity to the quality-gate strictness c (EER / FRR@FAR= 10^{-3} / FTE, %). A larger c rejects fewer images. At c=4 no image is rejected on most datasets. The main paper uses c=2 for all datasets.

Dataset	$c = 2$ (main)	$c = 3$	$c = 4$
V1	0.38 / 2.14 / 2.9	0.28 / 1.74 / 0.1	0.25 / 1.16 / 0.0
Int	0.22 / 0.76 / 3.5	0.25 / 0.69 / 0.2	0.19 / 0.49 / 0.0
Tho	0.22 / 0.54 / 0.1	0.22 / 0.56 / 0.0	0.22 / 0.56 / 0.0
Lam	0.00 / 0.00 / 4.0	0.00 / 0.00 / 0.5	0.00 / 0.00 / 0.0
Pol	0.04 / 0.02 / 3.5	0.01 / 0.00 / 0.0	0.01 / 0.00 / 0.0
MMU	1.04 / 6.47 / 4.0	0.83 / 5.99 / 1.8	0.83 / 6.23 / 0.9
Cog	0.60 / 4.21 / 0.0	0.60 / 4.21 / 0.0	0.60 / 4.21 / 0.0
Vis	0.28 / 0.83 / 3.3	0.31 / 0.84 / 0.1	0.28 / 0.70 / 0.0
EM	0.00 / 0.00 / 1.6	0.02 / 0.02 / 0.4	0.00 / 0.00 / 0.0
OED	0.57 / 31.32 / 0.4	0.58 / 30.27 / 0.0	0.58 / 30.27 / 0.0
Deg	0.25 / 1.23 / 3.1	0.27 / 1.74 / 0.2	0.29 / 2.00 / 0.0
Afr	0.10 / 0.20 / 6.3	0.14 / 0.62 / 3.1	0.11 / 0.27 / 1.7

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